

They Got Billions. We Got Data

Group Members: Tasnim & Tahana

Project Overview

If money doesn't buy happiness, why are billionaires still winning? Let's break it down. This project explores global billionaire wealth using a structured database built in DuckDB, focusing on three key questions:

1. **Where do billionaires come from?** We analyze which countries produce the most billionaires and examine whether a country's GDP plays a role in that.
2. **How is wealth created?** We compare self-made vs. inherited wealth across different regions and industries to see if opportunity or legacy matters more.
3. **Who becomes a billionaire?** We study differences in gender and age between self-made and inherited billionaires to understand who is most likely to build or inherit extreme wealth.

Simple idea: not just who is rich, but why they are rich.

Accessing data

```
#Read all three CSV files into R as data frames

billionaires_raw <- read.csv("billionaires.csv")
forbes_2023_raw  <- read.csv("Billionaires Statistics Dataset.csv")
countries        <- read.csv("Countries.csv")
```

Designing the database

```
# Delete any stale database file left over from a previous run.
# This is essential for reproducibility: if an old .duckdb file exists it
# will already contain tables created by dbWriteTable() (no DDL, no PRIMARY
# KEY), which causes two problems when we re-run:
#   1. CREATE TABLE fails with "table already exists"
#   2. FOREIGN KEY creation fails because the old table has no PK/UNIQUE
# Wiping the file guarantees a clean slate on every render.
db_path <- "billionaires_db.duckdb"
wal_path <- paste0(db_path, ".wal") # DuckDB write-ahead log (may not exist)

if (file.exists(db_path)) file.remove(db_path)
```

```
[1] TRUE
```

```
if (file.exists(wal_path)) file.remove(wal_path)
con <- dbConnect(duckdb::duckdb(), dbdir = db_path)
```

```
# Step 1: Rename columns and enforce R-level types before loading into DuckDB.
# Each field is explicitly selected, renamed to snake_case, and cast to the
# appropriate R type so that DuckDB inherits a clean, predictable schema.
```

```
billionaires <- billionaires_raw |>
  select(
    name,
    rank,
    year,
    company_name      = company.name,
    sector            = company.sector,
    company_relationship = company.relationship,
    company_type      = company.type,
    age               = demographics.age,
    gender            = demographics.gender,
    country           = location.citizenship,
    region            = location.region,
    country_gdp       = location.gdp,
    worth_billions    = wealth.worth.in.billions,
```

```

    inherited          = wealth.how.inherited,
    industry            = wealth.how.industry,
    was_founder         = wealth.how.was.founder, # "yes"/"no" string
    wealth_category     = wealth.how.category,
    from_emerging_market = wealth.how.from.emerging # "yes"/"no" string
  )|>
# Cast each column to its intended type so the DuckDB schema is accurate
mutate(
  rank          = as.integer(rank),
  year          = as.integer(year),
  age           = as.integer(age),
  country_gdp   = as.double(country_gdp),
  worth_billions = as.double(worth_billions),
  # Standardise free-text boolean flags to consistent lowercase strings
  was_founder   = tolower(trimws(was_founder)),
  from_emerging_market = tolower(trimws(from_emerging_market))
)

forbes_2023 <- forbes_2023_raw |>
select(
  rank,
  name          = personName,
  worth_billions_2023 = finalWorth, # raw value is in millions; kept as-is
  industry       = category,
  country,
  city,
  wealth_source  = source,
  industries,
  self_made      = selfMade, # logical TRUE/FALSE from R
  gender,
  age,
  citizenship_country = countryOfCitizenship
)|>
mutate(
  rank          = as.integer(rank),
  worth_billions_2023 = as.double(worth_billions_2023),
  age           = as.integer(age),
  self_made      = as.logical(self_made)
)

# countries was loaded directly from a lookup CSV; cast types explicitly
countries <- countries |>
mutate(

```

```

    gdp_per_capita = as.double(gdp_per_capita)
  ) |>
  distinct(country, .keep_all = TRUE)

```

```

# Step 2: Define the relational schema using explicit CREATE TABLE DDL.
#
# This is the key database-design step:
#   - Field types are declared (VARCHAR, INTEGER, DOUBLE, BOOLEAN) so DuckDB
#     stores data in the intended format, not whatever it infers from the CSV.
#   - NOT NULL constraints are applied to every field that must always have a
#     value for the row to be meaningful (primary identifiers, key measures).
#   - A PRIMARY KEY is set on countries.country, the anchor of the schema.
#   - FOREIGN KEY references document the join paths between tables, making
#     the relational design explicit and self-describing.
#
# Tables are dropped first so the chunk is idempotent (safe to re-run).
# Drop in dependency order (child tables before parent)
# Drop in dependency order (child tables before parent)
dbExecute(con, "DROP TABLE IF EXISTS billionaires")

```

[1] 0

```
dbExecute(con, "DROP TABLE IF EXISTS forbes_2023")
```

[1] 0

```
dbExecute(con, "DROP TABLE IF EXISTS countries")
```

[1] 0

```

# Parent / lookup table, one row per country
dbExecute(con, "
  CREATE TABLE countries (
    country      VARCHAR NOT NULL,    -- ISO country name; anchor for JOINS
    continent    VARCHAR,
    region       VARCHAR,
    gdp_per_capita DOUBLE,             -- World Bank estimate (USD)
    PRIMARY KEY (country)
  )
")

```

[1] 0

```
# Core historical billionaires table (CORGIS dataset, 1996-2014)
dbExecute(con, "
CREATE TABLE billionaires (
  name          VARCHAR NOT NULL, -- Billionaire's full name
  rank          INTEGER,          -- Forbes rank in the given year
  year          INTEGER NOT NULL, -- Snapshot year (1996, 2001, 2014)
  company_name  VARCHAR,
  sector        VARCHAR,
  company_relationship VARCHAR, -- e.g. founder, chairman, CEO
  company_type  VARCHAR,        -- e.g. private, public
  age           INTEGER,
  gender        VARCHAR,
  country       VARCHAR,        -- joins to countries.country
  region        VARCHAR,
  country_gdp   DOUBLE,          -- GDP embedded in source CSV
  worth_billions DOUBLE NOT NULL, -- Net worth in USD billions
  inherited     VARCHAR,        -- e.g. 'not inherited', 'father'
  industry      VARCHAR,
  was_founder   VARCHAR,        -- 'yes' / 'no' (string flag)
  wealth_category VARCHAR,
  from_emerging_market VARCHAR -- 'yes' / 'no' (string flag)
)
")
```

[1] 0

```
# 2023 Forbes snapshot (Kaggle dataset , single-year cross-section)
dbExecute(con, "
CREATE TABLE forbes_2023 (
  rank          INTEGER,
  name          VARCHAR NOT NULL, -- Billionaire's full name
  worth_billions_2023 DOUBLE,      -- Net worth in USD billions (2023)
  industry      VARCHAR,
  country       VARCHAR,          -- Country of residence
  city          VARCHAR,
  wealth_source VARCHAR,          -- Primary source of wealth
  industries     VARCHAR,          -- Broader industry categories
  self_made      BOOLEAN,          -- TRUE = self-made, FALSE = inherited
  gender        VARCHAR,
  age           INTEGER,

```

```

        citizenship_country VARCHAR NOT NULL    -- joins to countries.country
    )
")

```

```
[1] 0
```

```

# Step 3: Populate the tables using dbAppendTable().
# Because the schema is already defined above, dbAppendTable() inserts data
# into the pre-existing columns rather than inferring a new schema , this is
# the correct workflow when using explicit DDL.
dbAppendTable(con, "countries",    countries)
dbAppendTable(con, "billionaires", billionaires)
dbAppendTable(con, "forbes_2023",  forbes_2023)

```

```

# Step 4: Verify the schema and data load.
# DESCRIBE returns each column's name, type, nullability, key role, and
# default value, confirming that the DDL was applied as intended.
# Inspect column-level schema for each table
dbGetQuery(con, "DESCRIBE billionaires")

```

	column_name	column_type	null	key	default	extra
1	name	VARCHAR	NO	<NA>	<NA>	<NA>
2	rank	INTEGER	YES	<NA>	<NA>	<NA>
3	year	INTEGER	NO	<NA>	<NA>	<NA>
4	company_name	VARCHAR	YES	<NA>	<NA>	<NA>
5	sector	VARCHAR	YES	<NA>	<NA>	<NA>
6	company_relationship	VARCHAR	YES	<NA>	<NA>	<NA>
7	company_type	VARCHAR	YES	<NA>	<NA>	<NA>
8	age	INTEGER	YES	<NA>	<NA>	<NA>
9	gender	VARCHAR	YES	<NA>	<NA>	<NA>
10	country	VARCHAR	YES	<NA>	<NA>	<NA>
11	region	VARCHAR	YES	<NA>	<NA>	<NA>
12	country_gdp	DOUBLE	YES	<NA>	<NA>	<NA>
13	worth_billions	DOUBLE	NO	<NA>	<NA>	<NA>
14	inherited	VARCHAR	YES	<NA>	<NA>	<NA>
15	industry	VARCHAR	YES	<NA>	<NA>	<NA>
16	was_founder	VARCHAR	YES	<NA>	<NA>	<NA>
17	wealth_category	VARCHAR	YES	<NA>	<NA>	<NA>
18	from_emerging_market	VARCHAR	YES	<NA>	<NA>	<NA>

```
dbGetQuery(con, "DESCRIBE forbes_2023")
```

	column_name	column_type	null	key	default	extra
1	rank	INTEGER	YES	<NA>	<NA>	<NA>
2	name	VARCHAR	NO	<NA>	<NA>	<NA>
3	worth_billions_2023	DOUBLE	YES	<NA>	<NA>	<NA>
4	industry	VARCHAR	YES	<NA>	<NA>	<NA>
5	country	VARCHAR	YES	<NA>	<NA>	<NA>
6	city	VARCHAR	YES	<NA>	<NA>	<NA>
7	wealth_source	VARCHAR	YES	<NA>	<NA>	<NA>
8	industries	VARCHAR	YES	<NA>	<NA>	<NA>
9	self_made	BOOLEAN	YES	<NA>	<NA>	<NA>
10	gender	VARCHAR	YES	<NA>	<NA>	<NA>
11	age	INTEGER	YES	<NA>	<NA>	<NA>
12	citizenship_country	VARCHAR	NO	<NA>	<NA>	<NA>

```
dbGetQuery(con, "DESCRIBE countries")
```

	column_name	column_type	null	key	default	extra
1	country	VARCHAR	NO	PRI	<NA>	<NA>
2	continent	VARCHAR	YES	<NA>	<NA>	<NA>
3	region	VARCHAR	YES	<NA>	<NA>	<NA>
4	gdp_per_capita	DOUBLE	YES	<NA>	<NA>	<NA>

```
# Confirm row counts match the source data frames
```

```
dbGetQuery(con, "
  SELECT 'billionaires' AS table_name, COUNT(*) AS row_count FROM billionaires
  UNION ALL
  SELECT 'forbes_2023' AS table_name, COUNT(*) AS row_count FROM forbes_2023
  UNION ALL
  SELECT 'countries' AS table_name, COUNT(*) AS row_count FROM countries
")
```

	table_name	row_count
1	billionaires	2614
2	forbes_2023	2640
3	countries	197

```
#Preview the first 5 rows of each table
```

```
dbGetQuery(con, "SELECT * FROM billionaires LIMIT 5")
```

	name	rank	year	company_name	sector	company_relationship
1	Bill Gates	1	1996	Microsoft	Software	founder
2	Bill Gates	1	2001	Microsoft	Software	founder
3	Bill Gates	1	2014	Microsoft	Software	founder
4	Warren Buffett	2	1996	Berkshire Hathaway	Finance	founder
5	Warren Buffett	2	2001	Berkshire Hathaway	Finance	founder
	company_type	age	gender	country	region	country_gdp
1	new	40	male	United States	North America	8.10e+12
2	new	45	male	United States	North America	1.06e+13
3	new	58	male	United States	North America	0.00e+00
4	new	65	male	United States	North America	8.10e+12
5	new	70	male	United States	North America	1.06e+13
	worth_billions		inherited	industry	was_founder	wealth_category
1	18.5	not	inherited	Technology-Computer	true	New Sectors
2	58.7	not	inherited	Technology-Computer	true	New Sectors
3	76.0	not	inherited	Technology-Computer	true	New Sectors
4	15.0	not	inherited	Consumer	true	Traded Sectors
5	32.3	not	inherited	Consumer	true	Traded Sectors
	from_emerging_market					
1			true			
2			true			
3			true			
4			true			
5			true			

```
dbGetQuery(con, "SELECT * FROM forbes_2023 LIMIT 5")
```

	rank	name	worth_billions_2023	industry	
1	1	Bernard Arnault & family	211000	Fashion & Retail	
2	2	Elon Musk	180000	Automotive	
3	3	Jeff Bezos	114000	Technology	
4	4	Larry Ellison	107000	Technology	
5	5	Warren Buffett	106000	Finance & Investments	
	country	city	wealth_source	industries	self_made
1	France	Paris	LVMH	Fashion & Retail	FALSE
2	United States	Austin	Tesla, SpaceX	Automotive	TRUE
3	United States	Medina	Amazon	Technology	TRUE

4	United States	Lanai	Oracle	Technology	TRUE
5	United States	Omaha	Berkshire Hathaway	Finance & Investments	TRUE

	gender	age	citizenship	country
1	M	74		France
2	M	51		United States
3	M	59		United States
4	M	78		United States
5	M	92		United States

```
dbGetQuery(con, "SELECT * FROM countries LIMIT 5")
```

	country	continent	region	gdp_per_capita
1	Afghanistan	Asia	South Asia	502
2	Albania	Europe	Southern Europe	6494
3	Algeria	Africa	Northern Africa	3691
4	Andorra	Europe	Southern Europe	43047
5	Angola	Africa	Sub-Saharan Africa	2798

Relational Schema

The three tables are linked by the country column as a foreign key:

billionaires	countries	forbes_2023		name	country	FK
citizenship_country	country	FK	continent	name	region	FK
countries.country	worth_billions	gdp_per_capita	worth_billions_2023	inherited	self_made	
industry	industry	gender	gender	year	age	

Querying data using SQL

Query 1 : Which Countries Produce the Most Billionaires?

Some countries export oil, some export tech and some apparently export billionaires. This query counts how many billionaires each country has produced across all years in the dataset, then ranks them from highest to lowest. We use RANK() as a window function to assign a rank based on billionaire count.

```

query1 <- dbGetQuery(con, "
  SELECT
    country,
    COUNT(*)                               AS billionaire_count,
    ROUND(SUM(worth_billions), 1)          AS total_worth_billions,
    ROUND(AVG(worth_billions), 2)          AS avg_worth_billions,
    RANK() OVER (ORDER BY COUNT(*) DESC)   AS country_rank
  FROM billionaires
  WHERE country IS NOT NULL
    AND country != ''
  GROUP BY country
  ORDER BY billionaire_count DESC
  LIMIT 20
")
query1

```

	country	billionaire_count	total_worth_billions	avg_worth_billions
1	United States	903	3542.1	3.92
2	Germany	160	671.0	4.19
3	China	153	377.1	2.46
4	Russia	119	434.9	3.65
5	Japan	96	280.3	2.92
6	Brazil	81	224.7	2.77
7	Hong Kong	77	338.2	4.39
8	France	72	336.9	4.68
9	United Kingdom	65	190.6	2.93
10	India	63	210.8	3.35
11	Italy	58	220.3	3.80
12	Canada	53	175.3	3.31
13	Switzerland	51	168.6	3.31
14	Mexico	44	202.6	4.60
15	Taiwan	40	112.0	2.80
16	Spain	37	142.4	3.85
17	South Korea	36	86.8	2.41
18	Australia	33	93.2	2.82
19	Turkey	32	65.4	2.04
20	Indonesia	31	80.6	2.60

	country_rank
1	1
2	2
3	3
4	4

5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14
15	15
16	16
17	17
18	18
19	19
20	20

Finding: The United States isn't just winning , it's playing a completely different game. With 903 billionaires produced between 1996 and 2014, the US has nearly six times more than Germany in second place (160), and almost six times more than China in third (153). If billionaire production were a sport, everyone else would be competing for second place. But raw count isn't the whole story. Germany actually beats the US on average worth per billionaire, $4.19B$ vs $3.92B$, meaning German billionaires tend to be individually richer even though there are far fewer of them. France ($4.68B$) and Hong Kong ($4.39B$) show the same pattern: smaller clubs, wealthier members. China's position is particularly interesting. It ranks third in billionaire count but its average worth of $2.46B$ is one of the lowest in the top 10 , suggesting China produced a large volume of billionaires during this period who were, relatively speaking, just getting started. Given China's explosive economic growth after 2000, many of these fortunes were likely still being built when the dataset ends in 2014. At the bottom of the top 20, countries like Turkey ($2.04B$ average), Indonesia ($2.60B$), and South Korea ($2.41B$) show emerging economies beginning to generate serious wealth. The overall picture is clear: the United States dominates in volume, Europe dominates in individual wealth, and Asia is the story of rapid growth.

Query 2 : Self-Made vs. Inherited Wealth by Region

So after seeing which countries 'export' billionaires, the next question is, are they self-made, or just well-made by their families. This query breaks down billionaires by whether their wealth was inherited or self-made, grouped by world region. We calculate the average net worth and total count for each combination to see which regions produce more inherited vs. entrepreneurial wealth.

```

query2 <- dbGetQuery(con, "
SELECT
  region,
  inherited,
  COUNT(*)                               AS billionaire_count,
  ROUND(AVG(worth_billions), 2)          AS avg_worth_billions,
  ROUND(SUM(worth_billions), 1)          AS total_worth_billions,
  ROUND(
    COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (PARTITION BY region),
    1
  )                                       AS pct_of_region
FROM billionaires
WHERE region IS NOT NULL
AND region != '0'
  AND inherited IS NOT NULL
  AND inherited != ''
GROUP BY region, inherited
ORDER BY region, billionaire_count DESC
")

query2

```

	region	inherited	billionaire_count
1	East Asia	not inherited	409
2	East Asia	father	94
3	East Asia	3rd generation	20
4	East Asia	spouse/widow	7
5	East Asia	4th generation	3
6	East Asia	5th generation or longer	2
7	Europe	not inherited	409
8	Europe	father	144
9	Europe	3rd generation	62
10	Europe	4th generation	47
11	Europe	spouse/widow	18
12	Europe	5th generation or longer	18
13	Latin America	not inherited	89
14	Latin America	father	66
15	Latin America	3rd generation	16
16	Latin America	spouse/widow	6
17	Latin America	4th generation	4
18	Latin America	5th generation or longer	1
19	Middle East/North Africa	not inherited	67

20	Middle East/North Africa	father	43
21	Middle East/North Africa	3rd generation	6
22	Middle East/North Africa	spouse/widow	1
23	North America	not inherited	654
24	North America	father	195
25	North America	3rd generation	97
26	North America	spouse/widow	25
27	North America	4th generation	11
28	North America	5th generation or longer	10
29	South Asia	not inherited	46
30	South Asia	father	12
31	South Asia	3rd generation	6
32	South Asia	4th generation	3
33	South Asia	spouse/widow	2
34	Sub-Saharan Africa	not inherited	13
35	Sub-Saharan Africa	father	4
36	Sub-Saharan Africa	3rd generation	3

	avg_worth_billions	total_worth_billions	pct_of_region
1	2.83	1157.9	76.4
2	3.59	337.0	17.6
3	2.18	43.7	3.7
4	2.20	15.4	1.3
5	2.00	6.0	0.6
6	1.05	2.1	0.4
7	3.69	1510.9	58.6
8	4.49	646.2	20.6
9	3.74	231.6	8.9
10	2.60	122.1	6.7
11	5.01	90.1	2.6
12	2.84	51.1	2.6
13	3.57	317.5	48.9
14	2.77	182.5	36.3
15	2.11	33.7	8.8
16	4.40	26.4	3.3
17	3.73	14.9	2.2
18	1.10	1.1	0.5
19	2.87	192.2	57.3
20	2.62	112.6	36.8
21	2.28	13.7	5.1
22	1.10	1.1	0.9
23	3.66	2391.0	65.9
24	4.65	906.3	19.7
25	3.46	336.1	9.8

26	5.08	126.9	2.5
27	3.05	33.6	1.1
28	2.75	27.5	1.0
29	2.94	135.1	66.7
30	4.48	53.8	17.4
31	2.37	14.2	8.7
32	3.60	10.8	4.3
33	3.60	7.2	2.9
34	4.05	52.6	65.0
35	2.95	11.8	20.0
36	4.57	13.7	15.0

Finding: The ultimate question did they build it or just get the password to the family bank account? The data has answers, and they're surprisingly different depending on where in the world you look. North America is the land of the self-made. Nearly 66% of North American billionaires built their wealth from scratch, with 654 self-made billionaires generating 2,391*B* in total wealth. But here's the twist the 195 who inherited from their fathers average 4.65*B* each, compared to \$3.66*B* for the self-made crowd. Old money, it turns out, is still very good money. East Asia tells a hustle story. Over 76% of East Asian billionaires are self-made the highest entrepreneurial rate among major regions. China, Japan, and South Korea drove an era of first-generation wealth creation that the data captures clearly. The father-inherited group in East Asia averages 3.59*B*, notably higher than the self-made average of 2.83*B*, suggesting family businesses that were passed down tend to be the larger, more established ones. Europe is where old money lives. Europe has the most balanced split about 59% self-made and 41% inherited across various generations. What stands out is Europe's spouse/widow category, which averages 5.01*B* per person the highest of any group in any region. European dynastic wealth runs deep, and when it's passed through marriage it tends to be the largest fortunes of all. Third and fourth generation billionaires in Europe are also notably common compared to other regions, with 62 third-generation and 47 fourth-generation entries. The Middle East keeps it in the family. With 57% self-made and 37% inherited from fathers, the Middle East shows a strong family business culture. The relatively small number of third-generation or longer entries suggests that most of this wealth is still relatively young concentrated in the first or second generation. Latin America and South Asia lean self-made but inheritance runs close behind. Latin America sits at 49% self-made vs 36% father-inherited, the closest split of any region, suggesting a strong culture of family-run empires being passed down. South Asia tells a similar story with 67% self-made, but the 12 father-inherited billionaires average a hefty 4.48*B*, nearly 1.5*B* more than the self-made average. Sub-Saharan Africa is small but striking. Only 20 billionaires total, but 65% are self-made and they average 4.05*B* each, higher than the self-made average in North America. Africa's billionaire class is small, young, and building fast. The bottom line: self-made billionaires dominate every region of the world but inherited billionaires are almost always individually wealthier. Building it yourself gets you in the club, inheriting it gets you the VIP table. You can grind your way in but some people are born with reservations

Query 3 : Top 5 Industries by Total Billionaire Wealth

So now that we know who is rich let's figure out what they actually do. Time to answer the most important question did we choose the wrong major? This query identifies which industries have generated the most total billionaire wealth across the dataset. We use SUM to aggregate worth, then filter to the top 5 using LIMIT. We also include average worth and count to show whether an industry is dominant because of many billionaires or a few ultra-wealthy ones.

```
query3 <- dbGetQuery(con, "
SELECT
  industry,
  COUNT(*)                               AS billionaire_count,
  ROUND(SUM(worth_billions), 1)         AS total_worth_billions,
  ROUND(AVG(worth_billions), 2)        AS avg_worth_billions,
  ROUND(MAX(worth_billions), 2)        AS max_worth_billions
FROM billionaires
WHERE industry IS NOT NULL
  AND industry != ''
  AND industry != 'umbrella'
GROUP BY industry
ORDER BY total_worth_billions DESC
LIMIT 5
")
query3
```

	industry	billionaire_count	total_worth_billions	avg_worth_billions
1	Consumer	471	1756.3	3.73
2	Retail, Restaurant	281	1161.3	4.13
3	Technology-Computer	208	1015.2	4.88
4	Media	219	852.5	3.89
5	Real Estate	280	844.2	3.01
	max_worth_billions			
1	58.2			
2	64.0			
3	76.0			
4	72.0			
5	38.0			

Finding: If you want to become a billionaire, what major should we choose? The data from 1996–2014 gives a pretty clear answer.

Consumer goods is the undisputed champion. With 471 billionaires generating 1,756.3*B* in total wealth, this industry sits comfortably at the top. These are the people selling you food, clothing, and everyday products. It turns out that selling things billions of people buy every single day is an extremely reliable path to extreme wealth. Retail and restaurants are not far behind. 281 billionaires generated 1,161.3*B* in total wealth, with an average of 4.13*B* per person which is actually higher than consumer goods. This is the industry of people who figured out scale. Build the right model, replicate it everywhere, and the money follows. Technology is where the biggest individual fortunes live. Tech has fewer billionaires than consumer or retail, but its average worth of \$4.88*B* is the highest of any industry here. Tech does not just create lots of billionaires. It creates the wealthiest ones. Media is the quiet overachiever. 219 billionaires, 852.5*B* in total wealth, and a max worth of 72*B* which is second only to tech. Media moguls build empires that last decades, which explains why the fortunes at the very top are so large. Real estate is steady but not spectacular. It produces a lot of billionaires but the average worth of 3.01*B* is the lowest in this group. Real estate wealth builds slowly and steadily over time rather than through explosive growth. The simple takeaway: consumer goods and retail create the most billionaires. Technology creates the richest ones.

While we were lowkey stressing about whether it's too late to change our majors as rising seniors... turns out, we might actually be on the right path. So for the Upcoming freshmens our official recommendation is: Computer Science, Business, Finance, Marketing and, if possible, a minor in 'having rich parents.'

Query 4 : JOIN: Does a Country's GDP Predict Its Billionaire Count?

Okay, so now we're getting into the real economics question does being a rich country automatically mean you produce rich people? And I guess at some point all three of us (Tasnim, Tahana, and Professor Abhishek) have wondered should I just move to a richer country? I mean, that's kind of why we're here, right? :)

This is our JOIN query. We join the `billionaires` table with the `countries` lookup table on the country name to bring in GDP per capita and continent. This lets us ask: do wealthier countries (by GDP per capita) produce more billionaires, or is raw economic size what matters?

```
query4 <- dbGetQuery(con, "
SELECT
  b.country,
  c.continent,
  c.region,
  c.gdp_per_capita,
  COUNT(b.name)           AS billionaire_count,
  ROUND(SUM(b.worth_billions), 1) AS total_worth_billions,
  ROUND(AVG(b.worth_billions), 2) AS avg_worth_billions,
```



```

ROUND(
    COUNT(b.name) * 1.0 / (c.gdp_per_capita / 1000),
    4
)
AS billionaires_per_1k_gdp
FROM billionaires b
LEFT JOIN countries c
    ON LOWER(TRIM(b.country)) = LOWER(TRIM(c.country))
WHERE c.gdp_per_capita IS NOT NULL
    AND b.country IS NOT NULL
GROUP BY b.country, c.continent, c.region, c.gdp_per_capita
ORDER BY billionaire_count DESC
LIMIT 25
")
query4

```

	country	continent	region	gdp_per_capita
1	United States	Americas	North America	76399
2	Germany	Europe	Western Europe	51204
3	China	Asia	Eastern Asia	12556
4	Russia	Europe	Eastern Europe	12195
5	Japan	Asia	Eastern Asia	33815
6	Brazil	Americas	South America	8917
7	Hong Kong	Asia	Eastern Asia	49661
8	France	Europe	Western Europe	44408
9	United Kingdom	Europe	Northern Europe	46371
10	India	Asia	South Asia	2389
11	Italy	Europe	Southern Europe	34777
12	Canada	Americas	North America	54966
13	Switzerland	Europe	Western Europe	93720
14	Mexico	Americas	North America	10045
15	Taiwan	Asia	Eastern Asia	35513
16	Spain	Europe	Southern Europe	30104
17	South Korea	Asia	Eastern Asia	32423
18	Australia	Oceania	Australia and New Zealand	63529
19	Turkey	Asia	Western Asia	10674
20	Indonesia	Asia	South-Eastern Asia	4332
21	Malaysia	Asia	South-Eastern Asia	11109
22	Sweden	Europe	Northern Europe	61029
23	Singapore	Asia	South-Eastern Asia	82808
24	Israel	Asia	Western Asia	54658
25	Thailand	Asia	South-Eastern Asia	7808
	billionaire_count	total_worth_billions	avg_worth_billions	

1	903	3542.1	3.92
2	160	671.0	4.19
3	153	377.1	2.46
4	119	434.9	3.65
5	96	280.3	2.92
6	81	224.7	2.77
7	77	338.2	4.39
8	72	336.9	4.68
9	65	190.6	2.93
10	63	210.8	3.35
11	58	220.3	3.80
12	53	175.3	3.31
13	51	168.6	3.31
14	44	202.6	4.60
15	40	112.0	2.80
16	37	142.4	3.85
17	36	86.8	2.41
18	33	93.2	2.82
19	32	65.4	2.04
20	31	80.6	2.60
21	28	86.4	3.09
22	27	164.8	6.10
23	26	68.4	2.63
24	26	65.7	2.53
25	23	60.3	2.62

billionaires_per_1k_gdp

1	11.8195
2	3.1248
3	12.1854
4	9.7581
5	2.8390
6	9.0838
7	1.5505
8	1.6213
9	1.4017
10	26.3709
11	1.6678
12	0.9642
13	0.5442
14	4.3803
15	1.1263
16	1.2291
17	1.1103

18	0.5194
19	2.9979
20	7.1560
21	2.5205
22	0.4424
23	0.3140
24	0.4757
25	2.9457

Findings: Does living in a richer country mean more billionaires are produced there? The answer is yes, but with some very interesting exceptions. The United States makes the strongest case. With a GDP per capita of 76,399 and 903 billionaires generating 3,542.1B in total wealth, no country comes close. It is the clearest example of a high GDP country that also produces extreme individual wealth at scale. Good news for us, we choose the right country. But GDP alone does not tell the whole story. China has a GDP per capita of just 12,556 yet produced 153 billionaires. Russia is similar at 12,195 with 119 billionaires. Meanwhile Switzerland, one of the richest countries in the world at \$93,720 GDP per capita, produced only 51. High national income does not automatically mean more billionaires. Sweden is the most striking outlier. Only 27 billionaires, but an average worth of 6.10B each, the highest in the entire top 25. Swedish billionaires are rare, but when Sweden produces one, they tend to be extraordinarily wealthy. Singapore tells the opposite story. The highest GDP per capita in the dataset at 82,808, yet only 26 billionaires averaging 2.63B each. Singapore's wealth is broadly distributed across its population rather than concentrated at the very top. Emerging economies prove that opportunity beats income. India has a GDP per capita of just 2,389 yet produced 63 billionaires. Indonesia, Thailand, and Brazil tell similar stories not wealthy on average, but fast-growing enough to create pockets of extreme wealth. The bottom line: the countries that produce the most billionaires tend to be large economies with open markets. The countries that produce the wealthiest individual billionaires tend to be places where a single company can achieve global scale.

Query 5 : Gender Gap in Billionaire Wealth

Time to check, is the billionaire club inclusive or very exclusive. This query compares male vs. female billionaires across three dimensions: how many there are, how much they are worth on average, and what share of total billionaire wealth each gender controls. The wealth share column reveals the true scale of the gender gap beyond just headcount.

```
query5 <- dbGetQuery(con, "
  SELECT
    gender,
    COUNT(*)                               AS billionaire_count,
```

```

ROUND(AVG(worth_billions), 2)                AS avg_worth_billions,
ROUND(SUM(worth_billions), 1)                AS total_worth_billions,
ROUND(
    SUM(worth_billions) * 100.0 / SUM(SUM(worth_billions)) OVER (),
    2
)
)                AS pct_of_total_wealth,
ROUND(
    COUNT(*) * 100.0 / SUM(COUNT(*)) OVER (),
    2
)
)                AS pct_of_total_count
FROM billionaires
WHERE gender IS NOT NULL
    AND gender != ''
GROUP BY gender
ORDER BY billionaire_count DESC
")
query5

```

	gender	billionaire_count	avg_worth_billions	total_worth_billions
1	male	2328	3.52	8187.3
2	female	249	3.82	951.0
3	married couple	3	1.30	3.9
	pct_of_total_wealth	pct_of_total_count		
1	89.56	90.23		
2	10.40	9.65		
3	0.04	0.12		

Findings: The numbers here are simple. Out of 2,580 billionaires in the dataset, only 249 are women. That is less than 1 in 10. But count is not even the worst part. Men control 89.56% of all billionaire wealth, nearly 8,187.3*B* compared to women's 951*B*. There is one small twist worth noting. Female billionaires actually average slightly more per person, 3.82*B* versus 3.52*B* for men. The married couple category is almost too small to mention just 3 entries averaging 1.30*B* each but it is a reminder that some fortunes are genuinely shared and the dataset tries to capture that. The bottom line: 90% male, 90% of the wealth. The billionaire world between 1996 and 2014 was overwhelmingly a men's club, and the data leaves no room for debate on that point. So yeah the billionaire club is definitely exclusive just not in the way we'd hope. At this point, even if me and Tasnim joined the billionaire list, the stats would barely notice us.

Query 6 : Does Inheritance Affect When You Become a Billionaire?

So now we look at timing, does inheritance come with a fast-forward button? This query compares the average and median age of billionaires who inherited their wealth versus those who built it themselves. We use a CASE statement to simplify the `inherited` column (which has multiple values like “father”, “spouse”, “3rd generation”) into a clean binary: inherited vs. self-made.

```
query6 <- dbGetQuery(con, "
SELECT
  CASE
    WHEN inherited = 'not inherited' THEN 'self-made'
    ELSE 'inherited'
  END
  AS wealth_origin,
  inherited
  AS inherited_detail,
  COUNT(*)
  AS billionaire_count,
  ROUND(AVG(age), 1)
  AS avg_age,
  ROUND(MIN(age), 0)
  AS youngest,
  ROUND(MAX(age), 0)
  AS oldest,
  ROUND(AVG(worth_billions), 2)
  AS avg_worth_billions
FROM billionaires
WHERE age > 0
  AND inherited IS NOT NULL
  AND inherited != ''
GROUP BY wealth_origin, inherited_detail
ORDER BY wealth_origin, billionaire_count DESC
")

query6
```

	wealth_origin	inherited_detail	billionaire_count	avg_age	youngest
1	inherited	father	481	63.4	24
2	inherited	3rd generation	152	59.9	21
3	inherited	4th generation	61	59.5	28
4	inherited	spouse/widow	47	69.7	45
5	inherited	5th generation or longer	22	59.0	12
6	self-made	not inherited	1466	62.5	29
	oldest	avg_worth_billions			
1	95	4.17			
2	98	3.51			
3	88	2.87			
4	92	5.09			

5	85	3.18
6	96	3.60

Findings: If you inherit your wealth, do you get rich younger? You'd think so but not really. Self-made billionaires average 62.5 years old actually younger than those who inherited from a father (63.4) or a spouse/widow (69.7). The spouse/widow group is the oldest and the richest. At an average age of 69.7 and an average worth of \$5.09B, this group likely reflects older individuals who inherited late in life after the death of an already very wealthy partner. They did not build it and did not receive it young they inherited it at the finish line. The youngest billionaires come from the longest dynasties. The 5th generation or longer category has a youngest entry of just 12 years old someone who was born into so much wealth that they were already counted as a billionaire before they were a teenager. Self-made billionaires are the largest group by far, 1,466 people and their age range runs from 29 to 96, showing that building your own fortune is possible at almost any stage of life. The bottom line: inheritance does not make you significantly younger when you hit billionaire status. What it does is almost guarantee you get there. The only question is how much you start with.

Query 7 : How Has the Number of Billionaires Changed Over Time?

So while students like us were stressing about their stats assignments billionaires were multiplying. This query tracks billionaire count and total wealth year by year, revealing how the billionaire class has grown (or contracted) across different economic eras. We also calculate year-over-year growth using LAG() to compare each year to the previous one.

```
query7 <- dbGetQuery(con, "
SELECT
  year,
  COUNT(*)                                AS billionaire_count,
  ROUND(SUM(worth_billions), 1)           AS total_worth_billions,
  ROUND(AVG(worth_billions), 2)           AS avg_worth_billions,
  COUNT(*) - LAG(COUNT(*)) OVER (ORDER BY year) AS yoy_count_change,
  ROUND(
    (COUNT(*) - LAG(COUNT(*)) OVER (ORDER BY year))
    * 100.0 / LAG(COUNT(*)) OVER (ORDER BY year),
    1
  )                                       AS yoy_pct_change
FROM billionaires
WHERE year IS NOT NULL
GROUP BY year
ORDER BY year ASC
")
```

```
query7
```

	year	billionaire_count	total_worth_billions	avg_worth_billions
1	1996	423	1049.5	2.48
2	2001	538	1728.6	3.21
3	2014	1653	6454.4	3.90

	yoy_count_change	yoy_pct_change
1	NA	NA
2	115	27.2
3	1115	207.2

Findings: Meanwhile our bank accounts stayed the same, but billionaires clearly didn't. The billionaire class did not just grow between 1996 and 2014, it exploded. In 1996 there were 423 billionaires holding a combined 1,049.5B in wealth, with an average worth of 2.48B per person. Already an extraordinary concentration of wealth, but modest compared to what came next. By 2001 the count had jumped to 538 an increase of 115 billionaires, or 27.2% growth in just five years. Total wealth grew to 1,728.6B and the average worth per billionaire rose to 3.21B. By 2014 the numbers are staggering. 1,653 billionaires holding 6,454.4B in total wealth nearly four times the number from 1996 and more than six times the total wealth. That is an increase of 1,115 billionaires from 2001 alone, a 207.2% jump. The average worth per person also rose to \$3.90B, meaning not only were there far more billionaires but each one was individually richer too. So while people like us were refreshing our bank apps, they were refreshing the billionaire list One important note: this dataset only captures three snapshot years 1996, 2001, and 2014 rather than every year in between. The growth between 2001 and 2014 covers a 13 year span that includes the 2008 financial crisis, which temporarily wiped out many large fortunes before they recovered and surpassed previous highs.

Query 8 : JOIN: High GDP Countries With Surprisingly Few Billionaires

So now we flip the story , some countries are rich but not that rich. This is our second JOIN query. We join **billionaires** with **countries** to find nations that are economically wealthy (high GDP per capita) but have produced very few billionaires. These are the “underperformers” countries where national wealth has not translated into extreme individual wealth concentration. We define high GDP as above \$25,000 per capita and few billionaires as fewer than 10 across all years.

```
query8 <- dbGetQuery(con, "  
  SELECT  
    c.country,  
    c.continent,
```

```

    c.region,
    c.gdp_per_capita,
    COUNT(b.name) AS billionaire_count,
    ROUND(SUM(b.worth_billions), 1) AS total_worth_billions,
    ROUND(AVG(b.worth_billions), 2) AS avg_worth_billions
FROM countries c
LEFT JOIN billionaires b
    ON LOWER(TRIM(c.country)) = LOWER(TRIM(b.country))
GROUP BY c.country, c.continent, c.region, c.gdp_per_capita
HAVING c.gdp_per_capita > 25000
    AND COUNT(b.name) < 10
ORDER BY c.gdp_per_capita DESC
")

```

query8

	country	continent	region	gdp_per_capita
1	Monaco	Europe	Western Europe	234317
2	Liechtenstein	Europe	Western Europe	168146
3	Luxembourg	Europe	Western Europe	135605
4	Ireland	Europe	Northern Europe	102508
5	Qatar	Asia	Western Asia	83521
6	Iceland	Europe	Northern Europe	76078
7	Finland	Europe	Northern Europe	53655
8	San Marino	Europe	Southern Europe	50690
9	Belgium	Europe	Western Europe	50114
10	United Arab Emirates	Asia	Western Asia	49016
11	New Zealand	Oceania	Australia and New Zealand	48781
12	Macau	Asia	Eastern Asia	43903
13	Andorra	Europe	Southern Europe	43047
14	Bahamas	Americas	Caribbean	34732
15	Puerto Rico	Americas	Caribbean	34000
16	Malta	Europe	Southern Europe	33286
17	Kuwait	Asia	Western Asia	32095
18	Cyprus	Europe	Southern Europe	31552
19	Brunei	Asia	South-Eastern Asia	31449
20	Slovenia	Europe	Southern Europe	29418
21	Estonia	Europe	Northern Europe	28975
22	Czech Republic	Europe	Eastern Europe	27839
23	Lithuania	Europe	Northern Europe	25973
24	Bahrain	Asia	Western Asia	25660
	billionaire_count	total_worth_billions	avg_worth_billions	

1	3	4.6	1.53
2	2	2.9	1.45
3	0	NA	NA
4	8	30.5	3.81
5	0	NA	NA
6	0	NA	NA
7	4	6.6	1.65
8	0	NA	NA
9	4	9.0	2.25
10	5	16.5	3.30
11	3	10.8	3.60
12	2	2.8	1.40
13	0	NA	NA
14	0	NA	NA
15	0	NA	NA
16	0	NA	NA
17	7	15.5	2.21
18	4	19.7	4.93
19	0	NA	NA
20	0	NA	NA
21	0	NA	NA
22	6	18.4	3.07
23	1	1.0	1.00
24	1	1.0	1.00

Findings: Some of the richest countries in the world have almost no billionaires. That is not a mistake, it is one of the most interesting findings in this entire dataset. Monaco sits at the very top with a GDP per capita of 234,317, the highest in the dataset by a wide margin yet only 3 billionaires. Liechtenstein follows at 168,146 with just 2. Luxembourg, Qatar, Iceland and San Marino produced zero billionaires despite GDP per capita figures ranging from 76,000 to 135,000. These are genuinely wealthy nations where prosperity is broadly shared across the population rather than sitting at the extreme top. The most interesting case is Ireland. With a GDP per capita of 102,508 and 8 billionaires generating 30.5*B* in total wealth, Ireland comes closest to the threshold. The UAE and Kuwait stand out in the Middle East. Both have high GDPs driven by oil wealth, yet produced only 5 and 7 billionaires respectively. O The bottom line: high GDP per capita and high billionaire counts are not the same thing. Billionaires are a product of specific conditions large markets, private enterprise, and systems that allow wealth to concentrate at the very top. So maybe the goal isn't to live where billionaires are, but where you don't need to be one.

Creating data visualizations

Visualization 1 : World Choropleth: Billionaires Per Country

```
# Query: Aggregate total billionaire appearances per country across all years.
# COUNT(*) counts every row (one row = one billionaire-year observation), so
# a person appearing in 1996, 2001, and 2014 contributes 3 to their country.
# This gives a measure of sustained billionaire presence, not unique people.
v1_data <- dbGetQuery(con, "
  SELECT
    country,
    COUNT(*) AS billionaire_count
  FROM billionaires
  WHERE country IS NOT NULL AND country != ''
  GROUP BY country
  ORDER BY billionaire_count DESC
")
# Prepare the base world map polygon data from ggplot2's built-in map_data().
# ggplot2 uses its own country name conventions (e.g., "USA", "UK") which
# differ from our dataset's names , recode the mismatches so the join works.
world_map <- map_data("world") |>
  mutate(region = case_when(
    region == "USA" ~ "United States",
    region == "UK" ~ "United Kingdom",
    region == "South Korea" ~ "South Korea",
    region == "Czech Republic" ~ "Czech Republic",
    region == "Russia" ~ "Russia",
    TRUE ~ region # keep all other names as-is
  ))

map_data_joined <- world_map |>
  left_join(v1_data, by = c("region" = "country"))
# Build the choropleth:
# - geom_polygon() draws each country from its lat/long boundary points.
# - scale_fill_gradientn() applies a log10 colour scale so that low counts
#   (e.g., 1-5 billionaires) are still visually distinguishable from zero,
#   and the enormous US count doesn't wash out every other country.
# - coord_map("mollweide") uses an equal-area projection so larger countries
```

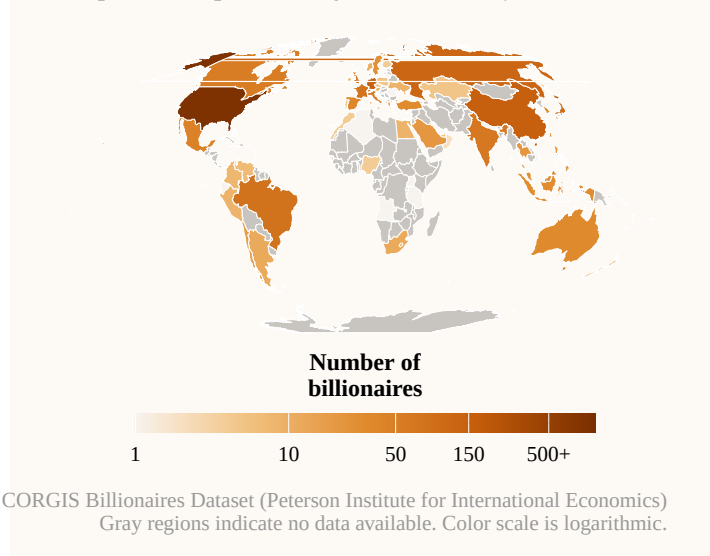
```

ggplot(map_data_joined, aes(x = long, y = lat, group = group, fill = billionaire_count)) +
  geom_polygon(color = "white", linewidth = 0.15) +
  scale_fill_gradientn(
    colors      = c("#f7f4f0", "#f0c27f", "#e08b2e", "#c45c0a", "#7a2e00"),
    na.value    = "#c8c4bf",      # neutral grey for countries with no data
    name        = "Number of\nbillionaires",
    trans       = "log10",      # log scale: prevents US from dominating colour range
    breaks      = c(1, 10, 50, 150, 500),
    labels      = c("1", "10", "50", "150", "500+"),
    limits      = c(1, 1000),
    oob         = scales::squish, # clamp any value > 1000 to the top colour
    guide       = guide_colorbar(
      barwidth      = 12,
      barheight     = 0.5,
      title.position = "top",
      title.hjust   = 0.5
    )
  ) +
  coord_map("mollweide") + # equal-area world projection
  labs(
    title      = "The Geography of Billionaire Wealth",
    subtitle   = "Total billionaires produced per country, 1996-2014 | Forbes World's Billiona
    caption    = "Source: CORGIS Billionaires Dataset (Peterson Institute for International Eco
  ) +
  theme_void(base_family = "serif") + # remove axes/grid lines , irrelevant on a map
  theme(
    plot.title      = element_text(size = 18, face = "bold", hjust = 0.5,
                                   margin = margin(b = 6)),
    plot.subtitle   = element_text(size = 11, color = "#666666", hjust = 0.5,
                                   margin = margin(b = 12)),
    plot.caption    = element_text(size = 8, color = "#999999", hjust = 1,
                                   margin = margin(t = 10)),
    legend.position = "bottom",
    legend.title    = element_text(size = 9, face = "bold"),
    legend.text     = element_text(size = 8),
    plot.margin     = margin(10, 20, 10, 20),
    plot.background = element_rect(fill = "#fdfaf6", color = NA)
  )

```

The Geography of Billionaire Wealth

Total billionaires produced per country, 1996–2014 | Forbes World's Billionaires



Interpretation: Alright, if the world map was a group project, some countries clearly did all the work. Billionaire wealth is not spread evenly across the world. The United States is the darkest point on the entire map and it is not close. With 500+ billionaires it sits in a category of its own, visually separating itself from every other nation on earth. Three regions dominate the rest of the map. Europe, East Asia, and parts of the Middle East all show medium to dark shading, reflecting strong billionaire counts in countries like Germany, China, Russia, Japan, and Saudi Arabia. Africa and Central Asia are almost entirely gray. Not light orange, gray, meaning no data or zero billionaires recorded. Across the entire African continent only a handful of countries register any color at all. The same is true for much of Central Asia and the Pacific. These regions were largely left out of the billionaire boom entirely during this period. Australia is a quiet standout. For a country of its size and population it shows a surprisingly solid shade, reflecting a small but established billionaire class built on mining, real estate, and retail. The bottom line: where billionaires come from is overwhelmingly a story about three parts of the world, North America, Europe, and East Asia. Everything else is the periphery. So, if anyone's planning to marry into billions these regions might be worth considering.

Visualization 2 : Lollipop Chart: Self-Made vs. Inherited by Region

```
# Query: Count billionaires per region × wealth origin.
# The CASE statement collapses the multi-value 'inherited' column into a
# clean binary label: anyone who is NOT 'not inherited' is treated as
# Inherited. This simplification is intentional, we want the high-level
```

```

# regional story, not a breakdown by generation of inheritance.
v2_data <- dbGetQuery(con, "
SELECT
  region,
  CASE
    WHEN inherited = 'not inherited' THEN 'Self-Made'
    ELSE 'Inherited'
  END AS wealth_origin,
  COUNT(*) AS billionaire_count,
  ROUND(AVG(worth_billions),2) AS avg_worth
FROM billionaires
WHERE region IS NOT NULL AND region != ''
  AND inherited IS NOT NULL AND inherited != ''
GROUP BY region, wealth_origin
")
# Reshape for the lollipop layout:
# 1. pivot_wider() creates one column per wealth_origin (Self-Made, Inherited),
#    so each row is a region with both counts side by side , needed to draw
#    the connecting segment between the two dots.
# 2. Rows with missing / blank region labels are removed.
# 3. A total column and % self-made are derived for ordering/annotation.
# 4. arrange(total) sorts regions so the chart reads low → high top → bottom.
v2_wide <- v2_data |>
  tidyr::pivot_wider(
    id_cols = region,
    names_from = wealth_origin,
    values_from = billionaire_count,
    values_fill = 0 ) |> # regions missing one category → 0, not NA

  filter(!is.na(region) & region != "" & region != "0") |>
  mutate(
    total = `Self-Made` + Inherited,
    pct_selfmade = round(`Self-Made` / total * 100, 1)
  ) |>
  arrange(total) # ascending so the smallest region is at the top of the chart
# Pivot back to long form for the geom_point() layers that need wealth_origin
v2_long <- v2_wide |>
  tidyr::pivot_longer(
    cols = c(`Self-Made`, Inherited),
    names_to = "wealth_origin",
    values_to = "count"
  ) |>

```

```

mutate(region = factor(region, levels = v2_wide$region))
# Build the lollipop chart:
# - geom_segment() draws the horizontal "stick" connecting Inherited (x)
#   to Self-Made (xend) for each region.
# - Two geom_point() layers place the orange (Inherited) and blue (Self-Made)
#   dots at their respective x positions.
# - geom_text() labels each dot with its count, nudged left/right to avoid
#   overlapping the dot itself.
# - annotate() layers add a manual legend (two dots + labels) because the
#   standard ggplot2 colour legend doesn't render cleanly for this geometry.
p2 <- ggplot(v2_wide, aes(y = factor(region, levels = v2_wide$region))) +
  geom_segment(
    aes(x = Inherited, xend = `Self-Made`, yend = region),
    color      = "#cccccc",
    linewidth  = 1.2 # thick enough to be visible but not distracting
  ) +
  geom_point(
    aes(x = Inherited),
    color = "#c45c0a", # warm orange = inherited wealth
    size  = 5,
    alpha = 0.9
  ) +
  geom_point(
    aes(x = `Self-Made`),
    color = "#2a6496", # cool blue = self-made wealth
    size  = 5,
    alpha = 0.9
  ) +
  geom_text(
    aes(x = Inherited, label = Inherited),
    nudge_x = -18,      # push label to the left of the orange dot
    nudge_y = 0,
    size    = 3,
    color   = "#c45c0a",
    fontface = "bold"
  ) +
  geom_text(
    aes(x = `Self-Made`, label = `Self-Made`),
    nudge_x = 18,       # push label to the right of the blue dot
    nudge_y = 0,
    size    = 3,
    color   = "#2a6496",

```

```

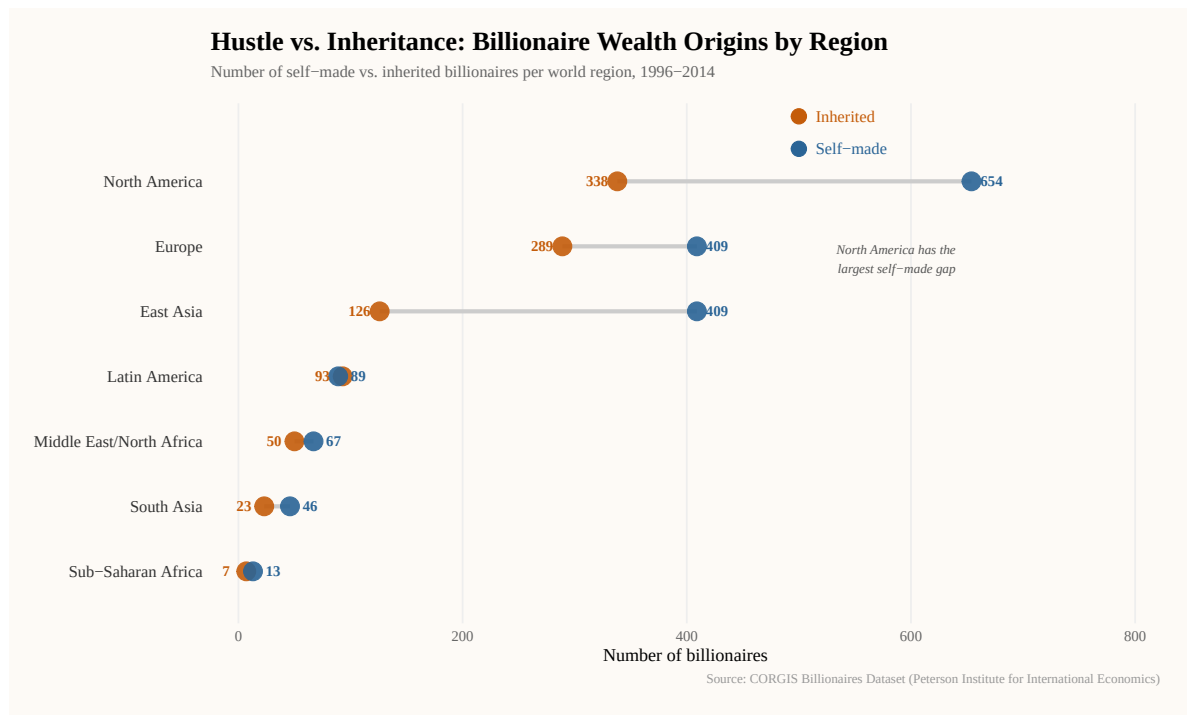
    fontface = "bold"
  ) +
  # Manual legend placed inside the plot area at a fixed coordinate
  annotate("point", x = 500, y = 8.0, color = "#c45c0a", size = 4) +
  annotate("text", x = 515, y = 8.0, label = "Inherited",
    hjust = 0, size = 3.5, color = "#c45c0a") +
  annotate("point", x = 500, y = 7.5, color = "#2a6496", size = 4) +
  annotate("text", x = 515, y = 7.5, label = "Self-made",
    hjust = 0, size = 3.5, color = "#2a6496") +
  # Annotation calling out the North America gap the widest in the chart
  annotate("text",
    x      = 640, y = 5.8,
    label  = "North America has the\nlargest self-made gap",
    hjust  = 1,
    size   = 2.8,
    color  = "#555555",
    fontface = "italic") +
  scale_x_continuous(
    expand = expansion(mult = c(0.02, 0.22)) # extra right margin for count labels
  ) +
  scale_y_discrete(
    expand = expansion(add = c(0.8, 1.2)) # breathing room above/below regions
  ) +
  labs(
    title      = "Hustle vs. Inheritance: Billionaire Wealth Origins by Region",
    subtitle   = "Number of self-made vs. inherited billionaires per world region, 1996-2014",
    x          = "Number of billionaires",
    y          = NULL,
    caption    = "Source: CORGIS Billionaires Dataset (Peterson Institute for International Ec
  ) +
  theme_minimal(base_family = "serif") +
  theme(
    plot.title      = element_text(size = 16, face = "bold", margin = margin(b = 6)),
    plot.subtitle   = element_text(size = 10, color = "#666666", margin = margin(b = 14)),
    plot.caption    = element_text(size = 8, color = "#999999", hjust = 1),
    axis.text.y     = element_text(size = 10, color = "#333333"),
    axis.text.x     = element_text(size = 9, color = "#666666"),
    panel.grid.major.y = element_blank(), # no horizontal gridlines the segments serve this
    panel.grid.major.x = element_line(color = "#eeeeee", linewidth = 0.5),
    panel.grid.minor  = element_blank(),
    plot.background  = element_rect(fill = "#fdfaf6", color = NA),
    plot.margin      = margin(15, 20, 20, 15)
  )

```

```

)
ggsave(
  "viz2_lollipop.png",
  plot    = p2,
  width   = 10,
  height  = 6,
  dpi     = 300,
  bg      = "#fdfaf6"
)
p2

```



Interpretation: Since this is a lollipop chart, let's see who actually earned their lollipop and who just got it handed to them by their family. It depends heavily on where in the world you look. North America is the clearest hustle story. The gap between the orange and blue dots is the widest of any region, 654 self-made billionaires versus 338 inherited. Nearly two thirds of North American billionaires built their fortune from scratch, which reflects the entrepreneurial culture the US is known for. East Asia matches North America's self-made count exactly, also 409, but from a much smaller inherited base of just 126. That means East Asia's billionaire class is even more heavily weighted toward first generation wealth builders, driven largely by China's economic rise and Japan and South Korea's industrial growth. Europe is the most balanced region. With 409 self-made and 289 inherited, Europe has the largest inherited billionaire class of any region. Old family money runs deep here and generational wealth

transfer is a well established tradition. Smaller regions tell a different story. Latin America, Middle East, South Asia, and Sub-Saharan Africa all have dots clustered close together near the left side of the chart , meaning small total numbers and relatively balanced splits between self-made and inherited. These regions are still early in their billionaire story. The bottom line: self-made billionaires outnumber inherited ones in every single region. But the size of that gap tells you a lot about each region's economic culture , wide gap means entrepreneurship, narrow gap means family wealth still plays a major role. So moral of the story if you can't build your own lollipop try being born into a candy shop.

Visualization 3 : Bar Chart: Top Industries by Total Wealth

```
library(forcats) # fct_reorder() for axis ordering
library(scales) # dollar_format(), comma() for axis labels
# Query: Pull top 10 industries by total accumulated wealth.
# 'umbrella' is filtered out : it's a catch-all category in the source data
# that doesn't correspond to a real industry and would skew the chart.
# avg_worth_billions is used as the fill aesthetic to show whether an
# industry's dominance is driven by many billionaires or by a few very rich ones.

v3_data <- dbGetQuery(con, "
  SELECT
    industry,
    COUNT(*) AS billionaire_count,
    ROUND(SUM(worth_billions), 1) AS total_worth_billions,
    ROUND(AVG(worth_billions), 2) AS avg_worth_billions
  FROM billionaires
  WHERE industry IS NOT NULL
    AND industry != ''
    AND industry != 'umbrella'
  GROUP BY industry
  ORDER BY total_worth_billions DESC
  LIMIT 10
")
# fct_reorder() sets the factor level order by total wealth so the longest
# bar is at the top , creates a naturally sorted horizontal bar chart.
v3_data <- v3_data |>
  mutate(industry = fct_reorder(industry, total_worth_billions))
# Midpoint of avg_worth range , used below to decide label colour (dark vs white)
# so that text is readable against both light and dark fill colours.
avg_midpoint <- (max(v3_data$avg_worth_billions) + min(v3_data$avg_worth_billions)) / 2
# Build the bar chart:
```

```

# - geom_col() fills by avg_worth_billions (a continuous gradient), encoding
#   two dimensions at once: bar length = total wealth, fill = avg wealth.
# - First geom_text() appends the total wealth figure to the right of each bar.
# - Second geom_text() overlays the billionaire count inside the bar; label
#   colour switches to white when the fill is dark (avg above midpoint).
# - scale_fill_gradientn() uses a sequential blue palette: light blue = lower
#   average wealth, dark navy = higher average wealth.
p3 <- ggplot(v3_data, aes(x = total_worth_billions, y = industry)) +
  geom_col(
    aes(fill = avg_worth_billions),
    width      = 0.65,
    show.legend = TRUE
  ) +
  # Label at the end of each bar: total wealth figure
  geom_text(
    aes(label = paste0("$", scales::comma(total_worth_billions), "B")),
    hjust    = -0.1,
    size     = 3.4,
    color    = "#333333",
    fontface = "bold"
  ) +
  # Label inside each bar: billionaire count, colour-coded for readability
  geom_text(
    aes(
      label = paste0(billionaire_count, " billionaires"),

      color = ifelse(avg_worth_billions >= avg_midpoint, "white", "#444444")
    ),
    hjust    = 1.08,
    size     = 3,
    fontface = "italic"
  ) +
  scale_color_identity() + # use the colour strings computed in aes() directly
  scale_fill_gradientn(
    colors = c("#aed6f1", "#2980b9", "#1a5276"), # light → dark blue gradient

    name    = "Avg. worth per\nbillionaire ($B)"
  ) +
  scale_x_continuous(
    labels = dollar_format(suffix = "B", prefix = "$"),
    expand = expansion(mult = c(0, 0.18)) # extra right-side space for end labels
  ) +

```

```

labs(
  title      = "Where Billionaire Wealth Is Built",
  subtitle   = "Top 10 industries by total accumulated wealth, 1996-2014",
  x          = "Total billionaire wealth (USD billions)"
) +
theme_minimal(base_family = "serif") +
theme(
  plot.title      = element_text(size = 16, face = "bold", margin = margin(b = 6)),
  plot.subtitle   = element_text(size = 10, color = "#666666", margin = margin(b = 14)),
  axis.title      = element_text(size = 10, color = "#444444"),
  axis.text       = element_text(size = 9, color = "#666666"),
  legend.position = "right",
  legend.margin   = margin(0, 0, 0, -10),
  legend.title    = element_text(size = 9, face = "bold"),
  legend.text     = element_text(size = 8),
  panel.grid.major = element_line(color = "#eeeeee", linewidth = 0.4),
  panel.grid.minor = element_blank(),
  plot.background = element_rect(fill = "#fdfaf6", color = NA),
  plot.margin     = margin(15, 60, 10, 15)
)
ggsave(
  "viz3_bars.png",
  plot      = p3,
  width     = 11,
  height    = 7,
  dpi       = 300,
  bg        = "#fdfaf6"
)

```

Interpretation: If this bar chart was a career fair, some industries would have very long lines. Not all industries are created equal when it comes to producing billionaires. The bar length tells you total wealth, the color tells you how rich each billionaire is on average, and together they reveal something interesting.

Consumer goods is the biggest industry by total wealth , 471 billionaires generating 1,756.3B. These are the people behind the food, drinks, and everyday products that billions of people buy without thinking. Volume is the game here and it pays extraordinarily well.

Technology has the darkest bar on the chart , meaning the highest average worth per person despite having fewer billionaires than consumer or retail. With 208 billionaires generating 1,015.2B, tech does not just create wealthy people. It creates the wealthiest ones. One successful platform can mint a fortune that takes a consumer goods company decades to match. Retail and restaurants sit comfortably in second place with 1,161.3B in total wealth across

281 billionaires. The formula is simple , find something that works and replicate it everywhere. Real Estate and Money Management both have light colored bars meaning lower average worth per billionaire despite decent total numbers. These industries build wealth steadily rather than explosively. You get rich, but rarely Silicon Valley rich. Diversified Financial stands out as a dark bar in the lower half. Only 167 billionaires but a noticeably darker shade , meaning finance billionaires tend to be individually wealthier than their numbers suggest.

The bottom line: if you want the most billionaires, sell everyday products. If you want the richest billionaires, build technology or get into finance. Lesson learned sell to everyone, scale it everywhere, or just invent the next app.

Visualization 4 : Scatter Plot: GDP Per Capita vs. Billionaire Count

```
library(ggrepel) # geom_label_repel() , non-overlapping country labels
# Query: JOIN countries (GDP) onto billionaires (counts) to enable the
# economic comparison. LEFT JOIN from countries so every country with GDP data
# is included even if it has 0 billionaires; HAVING filters to countries
# with at least one billionaire, since 0-count countries would cluster at the
# y-axis and clutter the chart without contributing insight.
# Point size encodes total wealth , a third dimension beyond x and y.
v4_data <- dbGetQuery(con, "
  SELECT
    c.country,
    c.continent,
    c.gdp_per_capita,
    COUNT(b.name) AS billionaire_count,
    ROUND(SUM(b.worth_billions), 1) AS total_worth_billions
  FROM countries c
  LEFT JOIN billionaires b
    ON LOWER(TRIM(c.country)) = LOWER(TRIM(b.country))
  GROUP BY c.country, c.continent, c.gdp_per_capita
  HAVING COUNT(b.name) > 0
  ORDER BY billionaire_count DESC
")
# Only label a curated set of notable / recognisable countries to avoid
# overplotting. Other points still appear as unlabelled dots.

label_countries <- c(
  "United States", "China", "Germany", "Russia",
  "India", "United Kingdom", "France", "Japan",
  "Switzerland", "Singapore", "Brazil"
```

```

)
# Add a label column: country name for selected countries, NA for all others.
# geom_label_repel() ignores NA labels automatically.
v4_labeled <- v4_data |>
  mutate(label = ifelse(country %in% label_countries, country, NA))
# Colour palette: one distinct colour per continent for easy group reading
continent_colors <- c(
  "Americas" = "#2a6496",
  "Europe"   = "#c45c0a",
  "Asia"     = "#27ae60",
  "Africa"   = "#8e44ad",
  "Oceania"  = "#c0392b"
)
# Build the scatter plot:
# - geom_smooth() fits a log-linear trend (y ~ log(x)) since both axes will
#   be log-scaled; the wide confidence band communicates high variability.
# - geom_point() maps continent to colour and total wealth to point size,
#   encoding three dimensions simultaneously.
# - geom_label_repel() draws non-overlapping callout labels for key countries.
# - annotate() adds a note on Singapore, the most interesting outlier.
# - Both axes are log-scaled because both GDP and billionaire counts span
#   several orders of magnitude; a linear scale would compress most countries
#   into an unreadable cluster at the low end.
p4 <- ggplot(v4_labeled,
  aes(x = gdp_per_capita, y = billionaire_count, color = continent)) +
  geom_smooth(
    method = "lm",
    formula = y ~ log(x), # log-linear fit matching the log-scaled axes
    se = TRUE, # show confidence interval ribbon
    color = "#999999",
    fill = "#dddddd",
    linewidth = 0.8,
    alpha = 0.4
  ) +
  geom_point(
    aes(size = total_worth_billions),
    alpha = 0.8,
    stroke = 0.4
  ) +
  geom_label_repel(
    aes(label = label),
    size = 3,

```

```

    fontface      = "italic",
    label.size     = 0.2,
    label.padding  = unit(0.2, "lines"),
    max.overlaps   = 15,
    show.legend    = FALSE,
    family         = "serif",

    box.padding    = unit(0.4, "lines"),
    point.padding  = unit(0.3, "lines"),
    min.segment.length = 0.2
  ) +
  # Annotation for Singapore high GDP but few billionaires (an outlier)
  annotate(
    "text",
    x      = 55000,
    y      = 22,
    label  = "Singapore: high GDP,\nfewer billionaires than expected",
    size   = 2.8,
    color  = "#555555",
    fontface = "italic",
    hjust  = 0
  ) +
  # Log-scale x axis: dollar labels in $Xk format for readability
  scale_x_log10(
    labels = function(x) paste0("$", scales::comma(x / 1000, accuracy = 1), "k"),
    breaks = c(1000, 5000, 10000, 25000, 50000, 100000)
  ) +
  scale_y_log10(
    labels = comma,
    breaks = c(1, 5, 10, 50, 100, 500)
  ) +
  scale_color_manual(
    values = continent_colors,
    name   = "Continent"
  ) +
  scale_size_continuous(
    range = c(2, 14), # smallest/largest point diameter in mm
    name  = "Total wealth\n(USD billions)",
    breaks = c(500, 1000, 2000, 3000),
    labels = dollar_format(suffix = "B", prefix = "$")
  ) +
  labs(

```

```

title      = "Does National Wealth Produce Billionaires?",

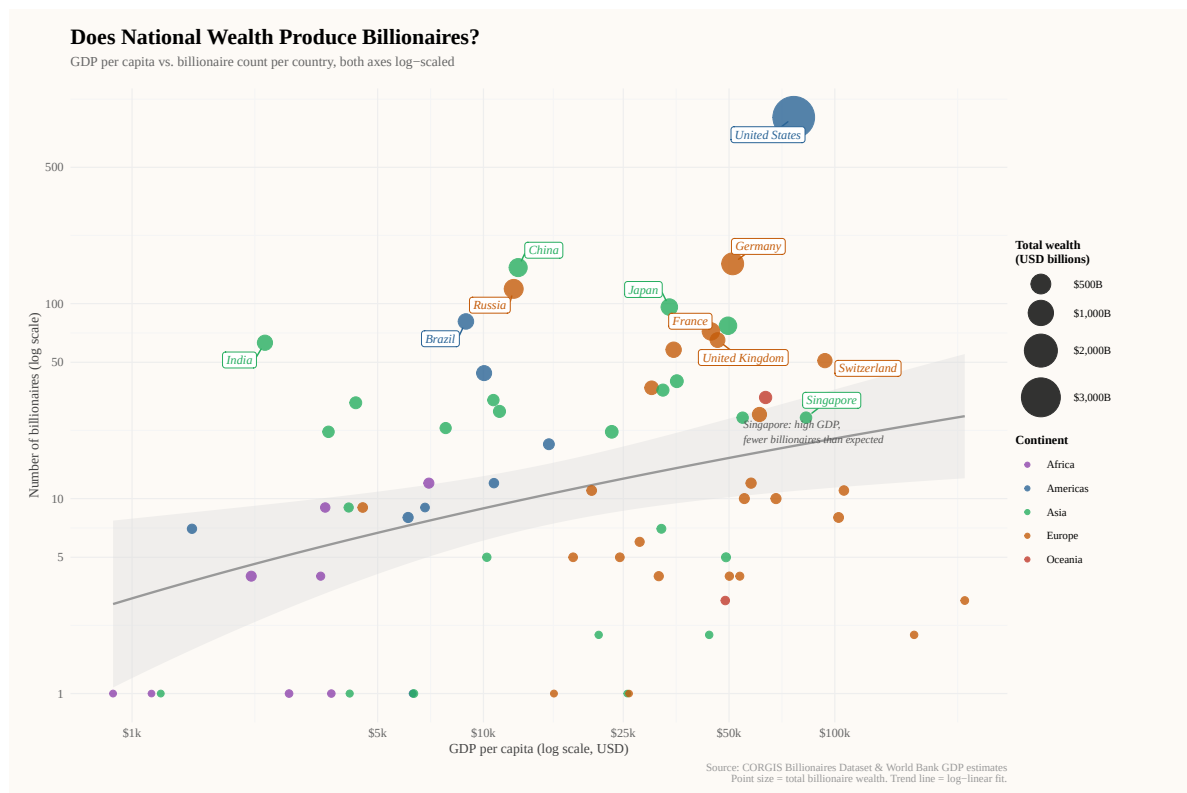
subtitle = "GDP per capita vs. billionaire count per country, both axes log-scaled",
x         = "GDP per capita (log scale, USD)",
y         = "Number of billionaires (log scale)",
caption   = "Source: CORGIS Billionaires Dataset & World Bank GDP estimates\nPoint size =
) +
theme_minimal(base_family = "serif") +
theme(
  plot.title      = element_text(size = 16, face = "bold",    margin = margin(b = 6)),
  plot.subtitle   = element_text(size = 10, color = "#666666", margin = margin(b = 14)),
  plot.caption    = element_text(size = 8,  color = "#999999", hjust = 1),
  axis.title      = element_text(size = 10, color = "#444444"),
  axis.text       = element_text(size = 9,  color = "#666666"),
  legend.position = "right",

  legend.margin   = margin(0, 0, 0, -5),
  legend.spacing.y = unit(0.3, "cm"),
  legend.title    = element_text(size = 9, face = "bold"),
  legend.text     = element_text(size = 8),
  panel.grid.major = element_line(color = "#eeeeee", linewidth = 0.4),
  panel.grid.minor = element_line(color = "#f5f5f5", linewidth = 0.2),
  plot.background = element_rect(fill = "#fdfaf6", color = NA),

  plot.margin     = margin(15, 60, 10, 15)
)

ggsave(
  "viz4_scatter.png",
  plot    = p4,
  width   = 12,
  height  = 8,
  dpi     = 300,
  bg      = "#fdfaf6"
)
p4

```



Interpretation: Does living in a rich country make you rich or are we just being delusional? This scatterplot answers it all. Richer countries tend to produce more billionaires, but the relationship is messier than you might expect, and the exceptions are where the real story lives.

The trend line confirms the general pattern. As GDP per capita rises, billionaire counts tend to rise too. But the wide confidence band around that line tells you the relationship is loose, plenty of countries break the rule in both directions.

The United States sits so far above the trend line it barely looks like part of the same chart. With a GDP per capita around \$76k and over 500 billionaires, no other country comes close. It is not just wealthy, it is structurally built to concentrate wealth at the very top in a way no other nation has matched.

China and India are the most impressive overperformers. Both sit well above the trend line despite relatively low GDP per capita, China at 12.5k and India at just 2.4k. These are countries where rapid economic growth created enormous first-generation fortunes even while the average citizen remained relatively poor.

Singapore is the chart's most thought-provoking dot. The highest GDP per capita on the right side of the chart, yet sitting right on or below the trend line with only around 26 billionaires.

European countries cluster neatly along the trend line , Germany, France, UK, Switzerland all behave exactly as the model predicts. Wealthy countries, solid billionaire counts, no surprises.

The bottom line: GDP per capita gets you started but it does not guarantee billionaires. What separates the overperformers is scale, speed of growth, and systems that allow wealth to concentrate fast. So even if we move, we might still be broke. Good to know.

Conclusion:

Finally, while GDP per capita is positively associated with billionaire presence, it does not fully explain it large markets, rapid economic growth, and systems that enable wealth concentration are more decisive factors. So, if you see us on the Forbes Billionaires List in a few years just remember it all started with this SQL project analysis or a lottery ticket, we're not sure yet.

```
# Close the DuckDB connection cleanly.  
# shutdown = TRUE ensures the WAL (write-ahead log) is flushed and the  
# .duckdb file is in a consistent, re-openable state.  
dbDisconnect(con, shutdown = TRUE)
```

Sources

- <https://corgis-edu.github.io/corgis/csv/billionaires/>
- <https://www.kaggle.com/datasets/nelgiryewithana/billionaires-statistics-dataset>
- LLM Prompt: Generate a small lookup table (~195 rows) with country, continent, region, and gdp_per_capita.